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Student Number



**Barker**  
College

**2020**

TRIAL HIGHER SCHOOL  
CERTIFICATE EXAMINATION

# Mathematics Extension 1

**Staff Involved:**

- ALY\*    • AXD
- JZT\*    • ESP
- ARM    • WMD

**FRIDAY 21<sup>st</sup> AUGUST**

100 copies

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**General**

**Instructions:**

- Reading time - 10 minutes
- Working time - 120 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and / or calculations

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**Total marks:**

**70**

**Section I - 10 marks (pages 2 - 5)**

- Attempt Multiple Choice Questions 1 - 10
- Allow about 15 minutes for this section

**Section II - 60 marks (pages 6 - 11)**

- Attempt Questions 11 - 14
- Show all necessary working
- Allow about 1 hour and 45 minutes for this section

## Section I

10 marks

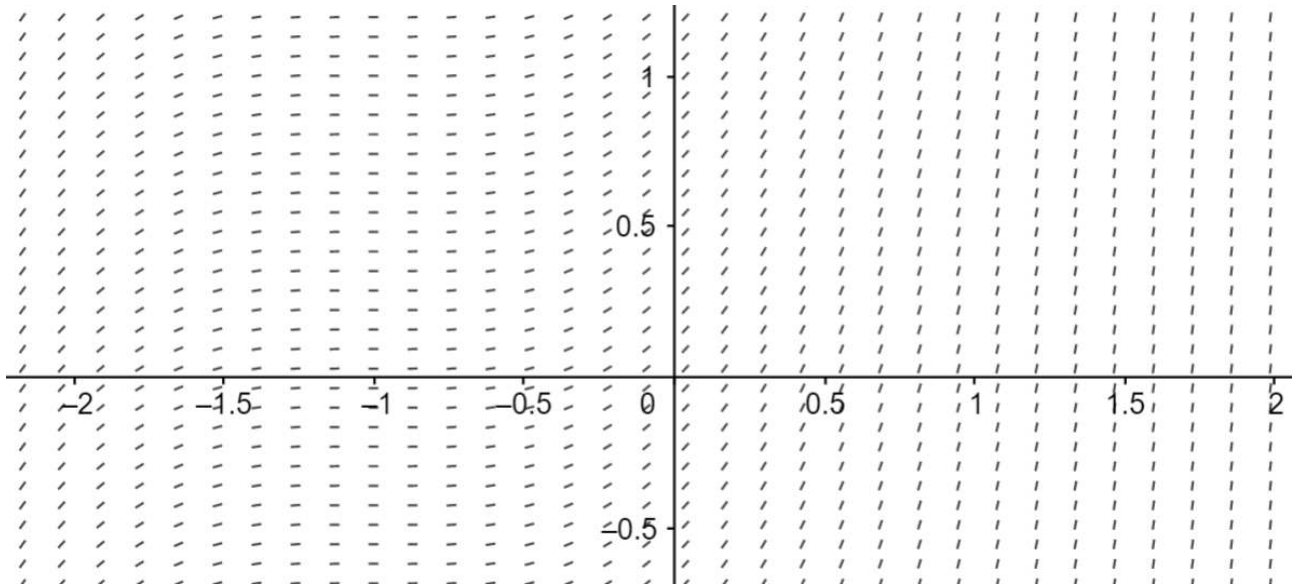
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

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1. Which differential equation would generate the direction field shown below?



(A)  $y = (x - 1)^2$

(B)  $y = (x + 1)^2$

(C)  $\frac{dy}{dx} = (x - 1)^2$

(D)  $\frac{dy}{dx} = (x + 1)^2$

2. Find the derivative of  $3 \sin^{-1}(2x)$

(A)  $\frac{3}{2\sqrt{1-4x^2}}$

(B)  $\frac{6}{\sqrt{1-4x^2}}$

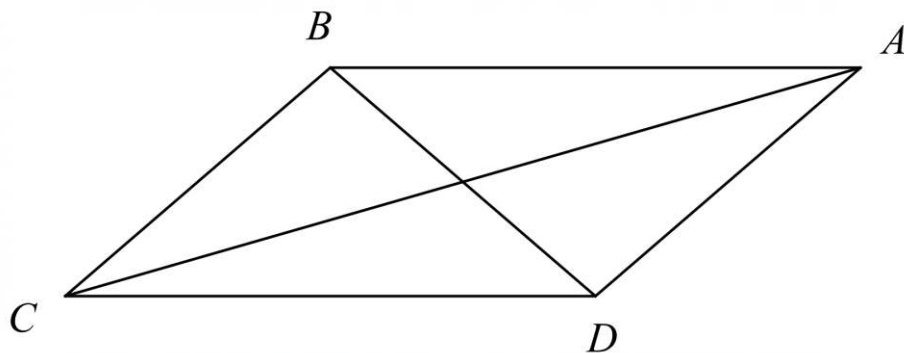
(C)  $\frac{3}{\sqrt{1-4x^2}}$

(D)  $\frac{12}{\sqrt{1-4x^2}}$

3. The polynomial  $P(x) = x^3 - 9x^2 - 21x + 245$  has a double root.  
The double root is:

(A)  $-7$   
(B)  $-5$   
(C)  $-1$   
(D)  $7$

4. If  $ABCD$  is a parallelogram, which vector is equal to  $\overrightarrow{AB} + \overrightarrow{BD}$  ?

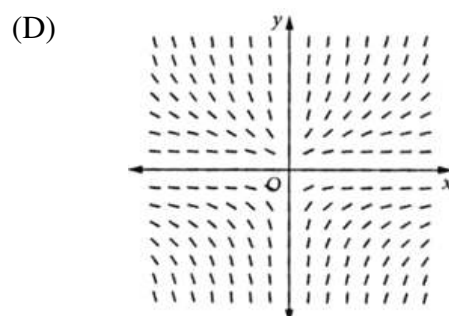
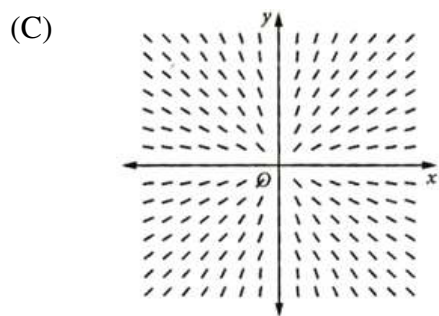
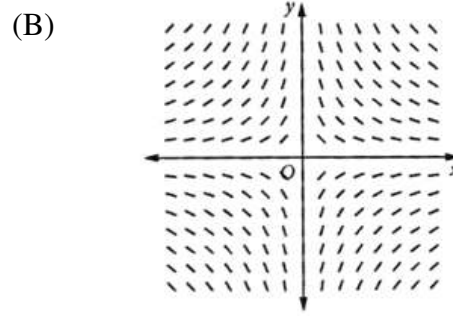
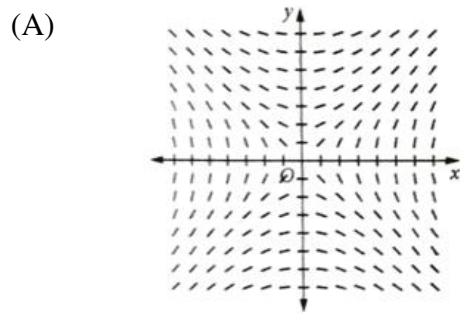


(A)  $\overrightarrow{DA}$   
(B)  $\overrightarrow{BC}$   
(C)  $\overrightarrow{AC}$   
(D)  $\overrightarrow{CD}$

5. In how many ways can 7 people be chosen from a group of 16 people and then arranged in a circle?

(A)  $\frac{15!}{9!}$   
(B)  $\frac{15!}{9!7!}$   
(C)  $\frac{16!}{9!}$   
(D)  $\frac{16!}{9!7}$

6. The slope field of  $xy' - y = 0$  could be



7. When the polynomial  $P(x)$  is divided by  $(x + 2)(x - 5)$  the remainder is  $3x - 6$ . What is the remainder when  $P(x)$  is divided by  $(x + 2)$ ?

- (A)  $-2$   
 (B)  $-7$   
 (C)  $-12$   
 (D)  $-28$

8. Using the substitution  $u = x^2 + 1$  or otherwise, evaluate  $\int_0^1 \frac{x}{(x^2 + 1)^2} dx$ .

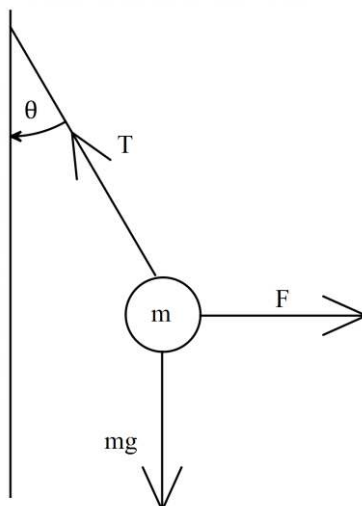
- (A)  $-\frac{1}{4}$   
 (B)  $-\frac{1}{2}$   
 (C)  $\frac{1}{4}$   
 (D)  $\frac{1}{2}$

9. Find the coefficient of  $x^5$  in the binomial expansion of

$$\left(\frac{1}{x^3} + 2x^2\right)^{15}$$

- (A) 96 096
- (B) 1 647 360
- (C) 2 562 560
- (D) 3 075 072

10. A mass of 4kg is attached to a wall by a light, inextensible string that makes an angle of  $\theta$  with the wall. A force,  $F$ , is applied to the mass away from the wall at an angle perpendicular to the wall. If the tension,  $T$ , in the string is 80N, what is the force  $F$  that is applied if the mass is in equilibrium? Assume acceleration due to gravity is  $10\text{ms}^{-2}$ .



- (A)  $40\sqrt{3}\text{N}$
- (B) 40N
- (C)  $40\sqrt{2}\text{N}$
- (D) 20N

**End of Section I**

## Section II

**60 marks**

**Attempt Questions 11 - 14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.  
Answer Question 11 (e) and (f) (ii), (iii) on the separate answer page provided.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

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| Question 11 (15 marks) | [START A NEW BOOKLET] | Marks |
|------------------------|-----------------------|-------|
| (a) Solve for $x$      |                       | 3     |

$$\frac{1}{x-2} > -1$$

|                                                                                                                   |   |
|-------------------------------------------------------------------------------------------------------------------|---|
| (b) Find the general solution of the differential equation $y' = \frac{-2x}{y}$ . Answer with $y$ as the subject. | 2 |
|-------------------------------------------------------------------------------------------------------------------|---|

|                                                                                               |   |
|-----------------------------------------------------------------------------------------------|---|
| (c) A particle moves so that its displacement $x$ from the origin is a function of time $t$ . | 2 |
|-----------------------------------------------------------------------------------------------|---|

Its velocity is given by the differential equation  $\frac{dx}{dt} = \sin^2 3t$ .

Initially, the particle is  $\frac{\pi}{4}$  metres to the right of the origin.

Find the particular solution to the differential equation that gives displacement as a function of time.

|              |   |
|--------------|---|
| (d) Evaluate | 2 |
|--------------|---|

$$\int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{1+4x^2}$$

**Answer Question 11 (e) on the separate answer page supplied.  
Hand it in with your booklet for Question 11.**

|                               |   |
|-------------------------------|---|
| (e) Sketch the slopefield for | 2 |
|-------------------------------|---|

$$y' = x - y.$$

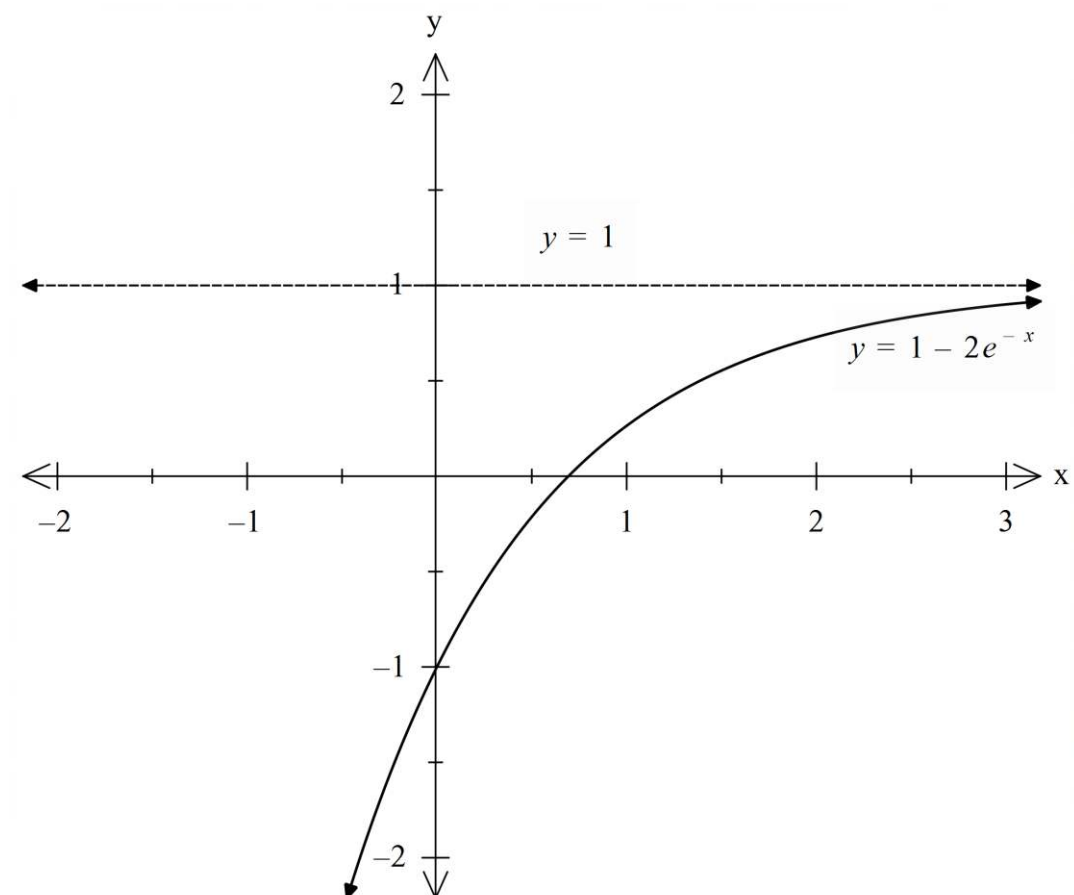
Draw one slope element at each of the points marked with a ' + '.

A table of slopes is also provided on the answer page if you wish to use it.

**Question 11 continues on the next page.**

**Question 11 continued.**

- (f) Consider the graph of  $f(x) = 1 - 2e^{-x}$  shown below.



- (i) Find the  $x$  intercept of this graph in exact form.

**1**

**Answer Question 11 (f) (ii) and (iii) on the separate answer page supplied.  
Hand it in with your booklet for Question 11.**

- (ii) Sketch  $y = f(|x|)$ . Show/label all important features.

**1**

- (iii) Sketch  $y = \frac{1}{f(x)}$ . Show/label all important features.

**2**

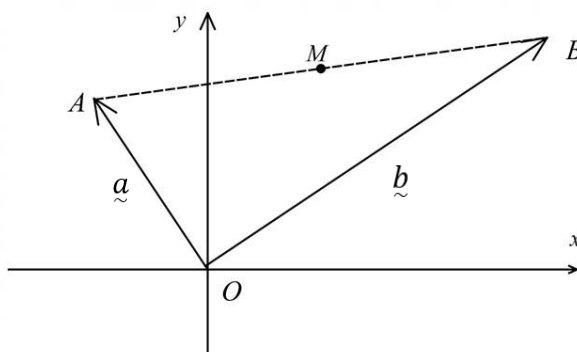
**End of Question 11**

**Question 12 (15 Marks)****[START A NEW BOOKLET]****Marks**

- (a) Evaluate  $x$  if the vectors  $\begin{pmatrix} x \\ 3 \end{pmatrix}$  and  $\begin{pmatrix} -6 \\ 8 \end{pmatrix}$  are:
- (i) perpendicular 1
- (ii) parallel 1

- (b) The mass  $M$  of a particular southern right whale is modelled as  $\frac{dM}{dt} = k(55 - M)$ , where  $M$  is measured in tonnes,  $t$  is the age of the whale in years, and  $k$  is a positive constant.
- (i) Solve the differential equation by integration to show that the general solution is  $M = 55 - Ae^{-kt}$ . 2
- (ii) This whale was 1 tonne at birth, and 10 tonnes when it was 1 year old. Find the whale's mass when it is 10 years old, to the nearest tonne. 3

- (c) Given  $\underline{a} \cdot \underline{b} = 0$ , show, with the aid of the diagram, that  $|\underline{a} + \underline{b}| = |\underline{a} - \underline{b}|$ , and hence show that the midpoint,  $M$ , of the hypotenuse of a right-angled triangle is equidistant from all three of its vertices. 3



- (d) (i) If  $t = \tan A$ , find an expression in terms of  $t$  for  $\cos 2A$ . 1
- (ii) Use  $t$  formulas with the substitution  $t = \tan x$  to prove that  $\sec 2x + \tan 2x = \tan(x + 45^\circ)$  2
- (iii) Hence or otherwise solve  $\sec 2x + \tan 2x = \frac{4}{5}$  for  $0^\circ \leq x \leq 360^\circ$  2
- Answer to the nearest degree.

**End of Question 12**

**Question 13 (15 marks)****[START A NEW BOOKLET]****Marks**

- (a) Consider the function  $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$
- (i) State the domain and range of  $f(x)$ . **1**
- (ii) Find and sketch the inverse function  $f^{-1}(x)$ . **3**

- (b) What is the least number of distinct integers that can be chosen from the sequence 1, 4, 7, 10, 13, 16, 19, ..., 97, 100 so that it is guaranteed that two of them will have a sum of 104? **2**

- (c) Prove by induction that **3**

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \cdots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}, \text{ for } n \geq 1$$

- (d) A ball rolling along a horizontal plane has position and velocity vectors given below:

$$\text{Position in metres: } \tilde{r} = x\tilde{i} + y\tilde{j}, \quad t \geq 0, \quad 0 < y < 1$$

$$\text{Velocity in } ms^{-1}: \dot{\tilde{r}} = 2y\sqrt{y}\tilde{i} + y(1-y)\tilde{j}$$

The component of the velocity in the  $\tilde{j}$  direction is the differential equation

$$\frac{dy}{dt} = y(1-y).$$

It can be shown that  $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$ . Do not prove this.

- (i) Solve the differential equation  $\frac{dy}{dt} = y(1-y)$  to find the general solution, and hence show that if the particle is initially at the point (0, 0.5), the component of the displacement in the  $\tilde{j}$  direction is: **3**

$$y = \frac{e^t}{1 + e^t}$$

- (ii) Show that the magnitude of the velocity vector is  $y(1+y)$  and hence calculate the magnitude of the velocity at time  $t = \ln 9$  seconds. **3**

**End of Question 13**

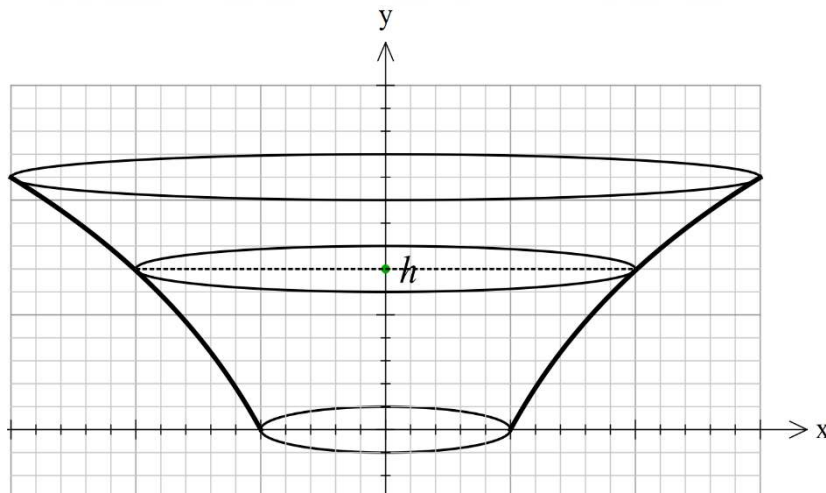
- (a) Consider the vectors  $\underline{a} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$  and  $\underline{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$ .

- (i) Show that  $\hat{\underline{b}} = \begin{pmatrix} \frac{-2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{pmatrix}$  1
- (ii) Hence, or otherwise, find  $\text{proj}_{\underline{b}} \underline{a}$  2
- (iii) Find the component of  $\underline{a}$  that is perpendicular to  $\underline{b}$ . 1

- (b) A bowl is formed by rotating the curve  $y = 2 \ln x$  around the  $y$  axis.

The volume of water in the bowl when the height of the water is  $h$ , can be modelled by rotating the area enclosed by the curve, the lines  $y = 0$  and  $y = h$ , about the  $y$  axis.

The lengths  $x$ ,  $y$  and  $h$  are in centimetres.



- (i) Show that the volume of water in the bowl is  $V = \pi(e^h - 1) \text{ cm}^3$ . 2
- (ii) A tap is dripping into this bowl at a rate of  $40 \text{ cm}^3/\text{minute}$ . 4

Find the rate at which the height of the water is increasing after 3 minutes, if the bowl is initially empty.

Give your answer in exact form.

**Question 14 continues on the next page.**

**Question 14 continued.**

- (c) (i) A ball is launched from ground level into the air with an initial speed of  $25 \text{ ms}^{-1}$  and just clears a 3 metre fence. The fence is 15 metres to the right of the launch site. **3**

You may assume the equations of motion for the ball:

$$x = 25t \cos \theta, \quad y = -4.9t^2 + 25t \sin \theta$$

By first finding the cartesian equation of motion, find the two possible angles of projection of the ball. Answer to the nearest **second**.

- (ii) For each launch angle, calculate the speed of the ball when it reaches the fence. **2**

Which launch angle gives the ball a greater speed when it reaches the fence?

**End of Paper**

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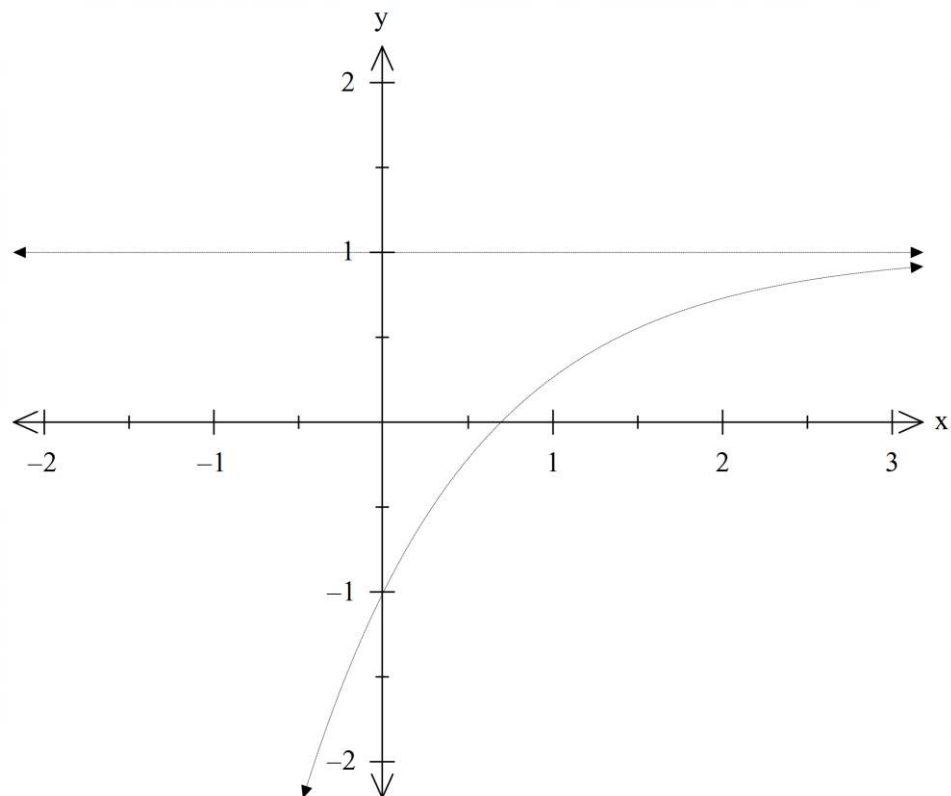
2

A blank coordinate plane with a grid. The x-axis is labeled 'x' in red at the bottom center. The y-axis is labeled 'y' in blue on the left side. The x-axis has tick marks and labels for -3, -2, -1, 0, 1, 2, and 3. The y-axis has tick marks and labels for -3, -2, -1, 0, 1, 2, and 3. The grid consists of 7 columns and 7 rows of squares.

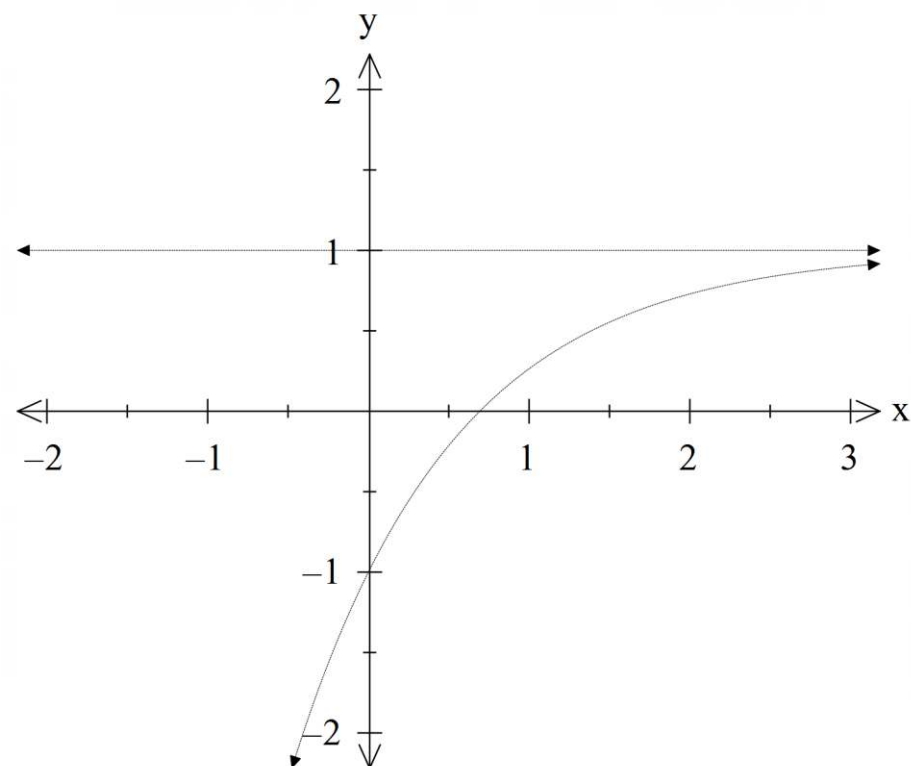
**Additional Answer page for Question 11 (f) (ii), (iii).**

**Answer Question 11(f) (ii) and (iii) here.**

11(f) (ii) Sketch on this answer page  $y = f(|x|)$ . Show/label all important features. 1



(iii) Sketch on this answer page  $y = \frac{1}{f(x)}$ . Show/label all important features. 2



1. D

2. B

$$\begin{aligned}\frac{d}{dx}(3\sin^{-1}2x) &= 3 \times 2 \times \frac{1}{\sqrt{1-(2x)^2}} \\ &= \frac{6}{\sqrt{1-4x^2}}\end{aligned}$$

3. D

$$\begin{aligned}P'(x) &= 3x^2 - 18x - 21 \\ &= 3(x+1)(x-7) \\ &= 0 \text{ when } x = -1 \text{ or } 7 \\ P(7) &= 0 \text{ and } P'(7) = 0 \\ \therefore \text{Double root is } 7\end{aligned}$$

4. B

$$\begin{aligned}\vec{AB} + \vec{BD} &= \vec{AD} \\ \vec{AD} &= \vec{BC} \text{ (opp sides, parallelogram)}\end{aligned}$$

5. D

$$\begin{aligned}{}^{16}C_7 \times 6! &= \frac{16!}{7!9!} \times 6! \\ &= \frac{16}{9!7}\end{aligned}$$

6. C

7. C

$$\begin{aligned}P(-2) &= 0 \times -7 + 3(-2) - 6 \\ &= -12\end{aligned}$$

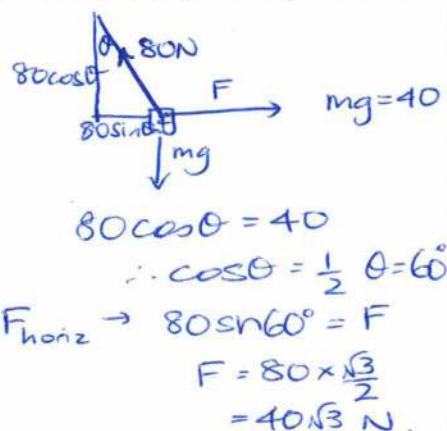
8. C

$$\begin{aligned}u &= x^2 + 1 & x=1 \ u=2 \\ & & x=0 \ u=1 \\ dy &= 2x \, dx \\ &= \frac{1}{2} \int \frac{1}{u^2} \\ &= -\frac{1}{2} \left[ \frac{1}{u} \right]_1^2 = -\frac{1}{2} \left( \frac{1}{2} - 1 \right) \\ &= \frac{1}{4}\end{aligned}$$

9. D

$$\begin{aligned}&\binom{15}{r} (x^{-3})^{15-r} (2x^2)^r \\ &= \binom{15}{r} x^{-45+3r+2r} 2^r \\ &-45+5r = 5 \\ &r = 10 \\ \text{coeff} &= \binom{15}{10} 2^{10} = 3075072\end{aligned}$$

10. A

Question 11

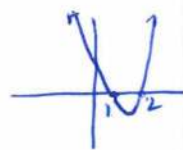
$$\text{a) } \frac{1}{x-2} > -1$$

$$(x-2) > -(x-2)^2$$

$$(x-2)^2 + (x-2) > 0$$

$$(x-2)(x-1) > 0$$

$$x < 1, x > 2$$



$$\text{b) } y' = \frac{-2x}{y}$$

$$\int y \, dy = -2 \int x \, dx$$

$$\frac{y^2}{2} = -\frac{2x^2}{2} + C$$

$$y^2 = -2x^2 + C$$

$$y = \pm \sqrt{-2x^2 + C}$$

$$\text{c) } \frac{dx}{dt} = \sin^2 3t$$

$$\int dx = \int \sin^2 3t \, dt$$

$$x = \frac{1}{2} \int (1 - \cos 6t) \, dt$$

$$= \frac{1}{2} \left[ t - \frac{1}{6} \sin 6t \right] + C$$

$$\text{at } t=0, x = \frac{\pi}{4}$$

$$\frac{\pi}{4} = \frac{1}{2} (0 - 0) + C \therefore C = \frac{\pi}{4}$$

$$\therefore x = \frac{1}{2} \left( t - \frac{1}{6} \sin 6t + \frac{\pi}{2} \right)$$

$$\text{d) } \int_0^{\frac{\sqrt{3}}{2}} \frac{dx}{1+4x^2} = \frac{1}{2} \int_0^{\frac{\sqrt{3}}{2}} \frac{2dx}{1+4x^2}$$

$$= \frac{1}{2} \left[ \tan^{-1} 2x \right]_0^{\frac{\sqrt{3}}{2}}$$

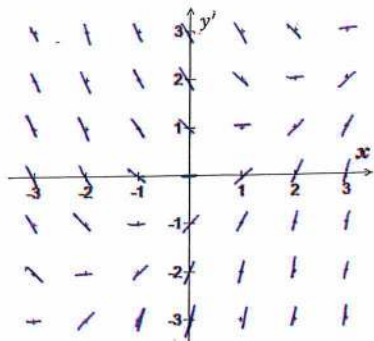
$$= \frac{1}{2} (\tan^{-1} \sqrt{3} - \tan^{-1} 0)$$

$$= \frac{\pi}{6}$$

ile

11. (e) On this answer page, sketch the slopefield for  $y' = x - y$ .  
Draw one slope element at each of the points marked with a '+'.  
A table of slopes is also provided if you wish to use it.

**Slope field:**

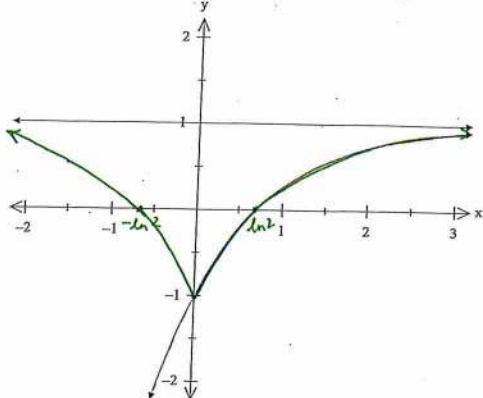


**Optional Table of Slopes:**

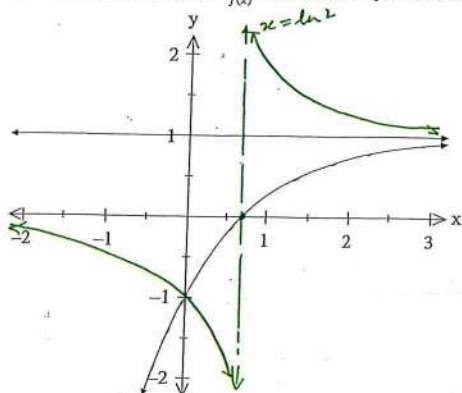
|    |    |    |    |    |    |    |   |
|----|----|----|----|----|----|----|---|
| 3  | -6 | -5 | -4 | -3 | -2 | -1 | 0 |
| 2  | -5 | -4 | -3 | -2 | -1 | 0  | 1 |
| 1  | -4 | -3 | -2 | -1 | 0  | 1  | 2 |
| 0  | -3 | -2 | -1 | 0  | 1  | 2  | 3 |
| -1 | -2 | -1 | 0  | 1  | 2  | 3  | 4 |
| -2 | -1 | 0  | 1  | 2  | 3  | 4  | 5 |
| -3 | 0  | 1  | 2  | 3  | 4  | 5  | 6 |
|    | -3 | -2 | -1 | 0  | 1  | 2  | 3 |

11f)

- (ii) Sketch on this answer page  $y = f(|x|)$ . Show/label all important features. 1



- (iii) Sketch on this answer page  $y = \frac{1}{f(x)}$ . Show/label all important features. 2



f)

- i) For  $x$ -int  
 $1 - 2e^{-x} = 0$   
 $e^{-x} = \frac{1}{2}$   
 $e^x = 2$

$$x = \ln 2.$$

### Question 12

12a (i) perp if  $\underline{a} \cdot \underline{b} = 0$

$$\begin{pmatrix} 7 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 8 \end{pmatrix}$$

$$-6x + 24 = 0$$

$$K = 4$$

ii) parallel if  $\underline{a} = k \underline{b}$

$$\begin{pmatrix} x \\ 3 \end{pmatrix} = k \begin{pmatrix} -6 \\ 8 \end{pmatrix}$$

$$x = -6k$$

and  $3 = 8k$

$$\therefore K = \frac{m}{\rho \times a}$$

$$x = -6 \times \frac{3}{8} = -\frac{9}{4}$$

$$12b) (i) \int \frac{dm}{55-m} = \int k dt$$

$$-\ln|55 - M| = kt + C$$

$$\ln(55 - M) = -kt + C,$$

$$e^{-kt+C_1} = |55 - M|$$

$$|55 - M| = e^{-kt} e^C$$

$$A = \pm e^C \quad 55 - M = Ae^{-kt}$$

$$M = 55 - Ae^{-kt} \text{ (shown)}$$

(ii)  $M = 55 - Ae^{-kt}$  at  $t=0, M=1$

$$I = 55 - A$$

$$\therefore A = 54$$

$$\therefore M = 55 - 54e^{-kt}$$

at  $t=1$   $M=10$

$$10 = 55 - 54e^{-k}$$

$$-45 = -54e^{-k}$$

$$e^{-k} = \frac{45}{54}$$

$$e^k = \frac{54}{45} \rightarrow k = \ln\left(\frac{54}{45}\right)$$

when  $t = 10$  <sup>ms</sup>

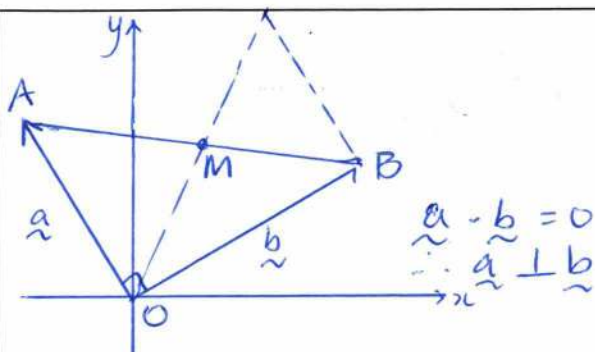
$$M = 55 - 54e^{-10 \cdot \ln\left(\frac{54}{45}\right)}$$

$$= 46.278 \dots$$

$= 46 \text{ tonnes (nearest tonne)}$

Solution

12c



$$\vec{BA} = -\vec{b} + \vec{a} = \vec{a} - \vec{b}$$

By Pythagoras

$$|\vec{a} - \vec{b}|^2 = |-\vec{b}|^2 + |\vec{a}|^2 \\ = |\vec{a}|^2 + |\vec{b}|^2$$

Construct  $\vec{BC}$  such that

$$\vec{BC} = \vec{OA} = \vec{a} \quad (\text{same mag. and direction}) \\ \therefore \vec{OC} = \vec{b} + \vec{a}$$

by Pythagoras

$$|\vec{OC}|^2 = |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \\ = |\vec{a} - \vec{b}|^2 \quad (\text{from above})$$

$$\therefore |\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2$$

$$\vec{BM} = \vec{MA} = \frac{1}{2}(\vec{a} - \vec{b})$$

(M is midpoint of BA)

$$|\vec{BM}| = |\vec{MA}| = \frac{1}{2}|\vec{a} - \vec{b}|$$

$$\vec{OM} = \vec{a} + \frac{1}{2}\vec{AM} = \vec{a} - \frac{1}{2}\vec{MA}$$

$$= \vec{a} - \frac{1}{2}(\vec{a} - \vec{b})$$

$$= \frac{1}{2}\vec{a} + \frac{1}{2}\vec{b}$$

$$\therefore |\vec{OM}| = \frac{1}{2}|\vec{a} + \vec{b}|$$

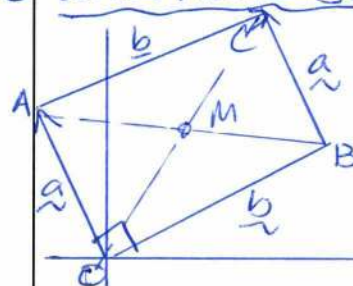
$$\therefore |\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2$$

$$\therefore |\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$$

$$\therefore |\vec{OM}| = |\vec{BM}| = |\vec{MA}|$$

12c

alternate solution.

Construct  $\vec{AC}$  parallel and equal in mag. to  $\vec{OB}$ 

$$\therefore \vec{AC} = \vec{OB} = \vec{b}$$

Similarly construct  $\vec{BC}$  such that  $\vec{BC} = \vec{OA} = \vec{a}$ 

$\therefore$  OBCA is a rectangle  
(opposite sides equal and parallel and 1 angle a right angle)

Since AB and OC are the diagonals of a rectangle, they bisect each other and are equal.

$$\therefore |\vec{AM}| = |\vec{BM}| = |\vec{OM}|$$

$$12d(i) \cos 2A = \frac{1-t^2}{1+t^2}$$

$$(ii) \text{ RTP } \sec 2x + \tan 2x = \tan(x+45^\circ)$$

$$\text{LHS} = \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2}$$

$$= \frac{t^2 + 2t + 1}{1-t^2}$$

$$= \frac{(t+1)^2}{(1-t)(1+t)}$$

$$= \frac{t+1}{1-t}$$

→

12d ii) cont'd

$$\begin{aligned} \text{RHS} &= \frac{\tan x + \tan 45^\circ}{1 - \tan x \tan 45^\circ} \\ &= \frac{t + 1}{1 - t} = \text{LHS} \\ &\therefore \text{proven.} \end{aligned}$$

12d (iii)  $\tan(x+45^\circ) = \frac{4}{5}$ 

$$0 \leq x \leq 360^\circ$$

$$45^\circ \leq x+45^\circ \leq 405^\circ$$

$$\text{ref acute} = 38.659^\circ$$

$$= 39^\circ \text{ (nearest deg)}$$

Quads 1 and 3

$$x+45^\circ = 180+39, 360+39^\circ$$

$$x = 174^\circ, 354^\circ$$

Question 1313a) (i) Domain of  $\sin^{-1}(x) = [-1, 1]$ so for  $3\sin^{-1}(\frac{x}{2})$ 

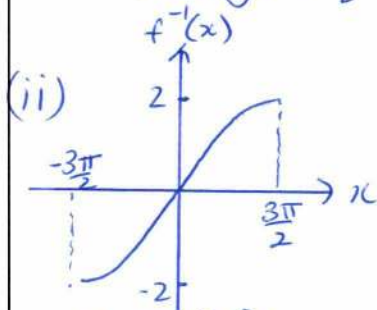
$$-1 \leq \frac{x}{2} \leq 1$$

$$\text{Domain} = [-2, 2]$$

$$\text{Range of } \sin^{-1}(x) = [-\frac{\pi}{2}, \frac{\pi}{2}]$$

so for  $3\sin^{-1}(\frac{x}{2})$ 

$$\text{Range} = [-\frac{3\pi}{2}, \frac{3\pi}{2}]$$



$$y = 3\sin^{-1}(\frac{x}{2})$$

$$f^{-1}(x) \rightarrow x = 3\sin^{-1}(\frac{y}{3})$$

$$\frac{y}{3} = \sin(\frac{x}{3})$$

$$f^{-1}(x) = 2\sin(\frac{x}{3})$$

13b) ① 4, 7, 10, 13, ..., 94, 97, 100

Arth series  $a=1$   $d=3$   $T_n=100$ 

$$a+(n-1)d = 100$$

$$1+3n-3 = 100$$

$$n = 34$$

 $\therefore 34$  values16 pairs add to 104  
+ number 1 + an extra  
in the middle. $\therefore$  Could choose 16 digits  
with no pair + 2 digits  
(1 and extra) = 18 total $\therefore$  Min is 19 by PHP.

$$13c) \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!} = 1 - \frac{1}{(n+1)!}$$

① Prove for  $n=1$ 

$$\text{LHS} = \frac{1}{2!} = \frac{1}{2}$$

$$\text{RHS} = 1 - \frac{1}{(1+1)!} = 1 - \frac{1}{2} = \frac{1}{2}$$

 $\therefore$  true for  $n=1$ ② Assume for  $n=k$ 

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

③ Prove true for  $n=k+1$ 

$$\text{RTP } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{(k+1)}{(k+2)!}$$

$$= 1 - \frac{1}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

$$\text{LHS} = \left( \frac{1}{2!} + \frac{2}{3!} + \dots + \frac{k}{(k+1)!} \right) + \frac{(k+1)}{(k+2)!}$$

$$= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \text{ by assumption}$$

$$= \frac{(k+1)! - 1}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= \frac{(k+2)! - (k+2) + k+1}{(k+2)!} \rightarrow$$

13c) cont'd  

$$\text{LHS} = \frac{(k+2)! - k+2+k+1}{(k+2)!}$$

$$= \frac{(k+2)! - 1}{(k+2)!}$$

$$= \text{RHS} \quad \therefore \text{True for } n=k+1$$

$$\therefore \text{Proven by MI}$$

13d)  $\underline{r} = x\underline{i} + y\underline{j}$   
 $\underline{\dot{r}} = 2y\sqrt{y}\underline{i} + y(1-y)\underline{j}$

(i)  $\frac{dy}{dt} = y(1-y)$

$$\int \frac{dy}{y(1-y)} = \int dt$$

$$\int \left( \frac{1}{y} + \frac{1}{1-y} \right) dy = t + C$$

$$\ln|y| - \ln|1-y| = t + C$$

$$\ln \left| \frac{y}{1-y} \right| = t + C$$

$$e^{t+C} = \left| \frac{y}{1-y} \right|$$

Let  $A = \pm e^C$

$$Ae^t = \frac{y}{1-y} \quad \text{at } t=0, y=0.5$$

$$A = \frac{0.5}{1-0.5} = 1$$

$$e^t = \frac{y}{1-y}$$

$$y = e^t - ye^t$$

$$y(1+e^t) = e^t$$

$$y = \frac{e^t}{1+e^t}$$

13d(ii)  $|\underline{\dot{r}}| = \sqrt{(2y\sqrt{y})^2 + y^2(1-y)^2}$   

$$= \sqrt{4y^3 + y^2 - 2y^3 + y^4}$$
  

$$= \sqrt{y^4 + 2y^3 + y^2}$$

$$= y\sqrt{y^2 + 2y + 1}$$

$$= y\sqrt{(y+1)^2}$$

$$= y(y+1)$$

at  $t = \ln 9$   $y = \frac{9}{1+9} = 0.9$

$$\therefore |\underline{\dot{r}}(0.9)| = 0.9(1.9) = 1.71$$

### Question 14

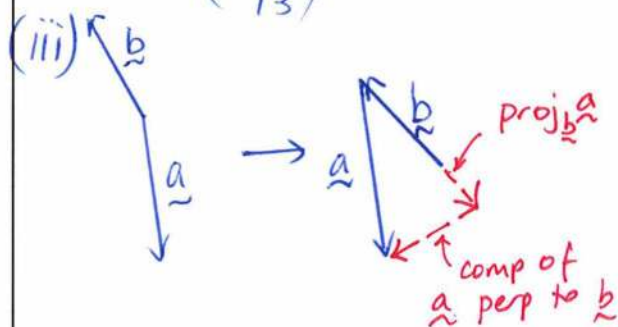
a) (i)  $\hat{\underline{b}} = \frac{\underline{b}}{|\underline{b}|} = \frac{1}{\sqrt{(-2)^2 + 3^2}} \times \begin{pmatrix} -2 \\ 3 \end{pmatrix}$   

$$= \begin{pmatrix} \frac{-2}{\sqrt{13}} \\ \frac{3}{\sqrt{13}} \end{pmatrix}$$

(ii)  $\text{proj}_{\hat{\underline{b}}} \underline{a} = \frac{\underline{a} \cdot \hat{\underline{b}}}{|\hat{\underline{b}}|} \hat{\underline{b}}$   

$$= \frac{-4 - 15}{\sqrt{13} \times \sqrt{13}} \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$
  

$$= \begin{pmatrix} \frac{38}{13} \\ -\frac{57}{13} \end{pmatrix}$$



Component  $= -\text{proj}_{\hat{\underline{b}}} \underline{a} + \underline{a}$   

$$= -\begin{pmatrix} \frac{38}{13} \\ -\frac{57}{13} \end{pmatrix} + \begin{pmatrix} 2 \\ -5 \end{pmatrix}$$
  

$$= \begin{pmatrix} -\frac{12}{13} \\ -\frac{8}{13} \end{pmatrix}$$

14b (i)  $\frac{y}{2} = \ln x \rightarrow x = e^{\frac{y}{2}}$

$$x^2 = e^y$$

$$V = \pi \int_0^h e^y dy$$

$$= \pi [e^y]_0^h$$

$$= \pi (e^h - 1) \text{ cm}^3$$

(ii)  $\frac{dh}{dt} = ? \quad \frac{dV}{dt} = 40 \quad \frac{dV}{dh} = \pi e^h$   
(from above)

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{1}{\pi e^h} \times 40 = \frac{40}{\pi e^h}$$

at  $t = 3 \quad V = 120$

$$120 = \pi (e^h - 1)$$

$$e^h = \frac{120}{\pi} + 1$$

$$\therefore \frac{dh}{dt} = \frac{40}{\pi \left( \frac{120}{\pi} + 1 \right)}$$

$$= \frac{40}{120 + \pi} \text{ cm/min}$$

14c (i)  $x = 25t \cos \theta$

$$\therefore t = \frac{x}{25 \cos \theta} \text{ sub into } y$$

$$y = -4.9 \left( \frac{x^2}{25^2 \cos^2 \theta} \right) + 25 \left( \frac{x}{25 \sin \theta} \right) \sin \theta$$

$$= -\frac{4.9}{625} x^2 \sec^2 \theta + x \tan \theta$$

$$= -0.00784 x^2 (1 + \tan^2 \theta) + x \tan \theta$$

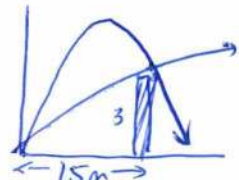
$$= -0.00784 x^2 - 0.00784 \tan^2 \theta + x \tan \theta$$

at  $x = 15, y = 3$

$$1.76 \tan^2 \theta - 15 \tan \theta + 4.764 = 0$$

$$\tan \theta = \frac{15 \pm \sqrt{225 - 4 \times 4.764 \times 1.764}}{2 \times 1.764}$$

$$\theta = 83^\circ 1' 28'' \text{ or } 18^\circ 17' 8''$$

14c (ii)   $\dot{x} = 25 \cos \theta$   
 $\dot{y} = -9.8t + 25 \sin \theta$

at  $x = 15, \theta = 83^\circ 1' 28''$

$$t = \frac{15}{25 \cos(83^\circ 1' 28'')}$$

$$\dot{x} = 25 \cos(83^\circ 1' 28'')$$

$$\dot{y} = -9.8 \left( \frac{3}{5 \cos \theta} \right) + 25 \sin \theta$$

$$= -23.60167812$$

$$\text{speed} = |\underline{v}| = \sqrt{\dot{x}_1^2 + \dot{y}_1^2}$$

$$= 23.79616351 \text{ ms}^{-1}$$

$$= 23.8 \text{ ms}^{-1} (1 \text{ dp})$$

at  $x = 15 \quad \theta = 18^\circ 17' 8''$

$$\dot{x}_2 = 25 \cos(18^\circ 17' 8'')$$

$$\dot{y}_2 = -9.8 \left( \frac{3}{5 \cos \theta} \right) + 25 \sin \theta$$

$$= 1.651124345$$

$$|\underline{v}| = \sqrt{\dot{x}_2^2 + \dot{y}_2^2}$$

$$= 23.794$$

$$= 23.8 \text{ ms}^{-1} (1 \text{ dp})$$

$\therefore$  Both speeds are equal at that point.